Exercises in Marine Ecological Genetics

10. Metabarcoding / eDNA I

- Amplicon Sequence Variants (ASVs)
- Taxonomic assignment
- ASV abundance distributions

Martin Helmkampf

https://github.com/mhelmkampf/meg26

Today's objectives

By the end of this session, you will be able to:

- Inspect **Amplicon Sequence Variant (ASV) count tables**, the backbone of eDNA data
- **Classify ASVs taxonomically** using BLAST against a public reference database
- **Construct a phyloseq object** by combining ASV counts, taxonomic assignments, and sample metadata, then filter low-abundance samples and ASVs
- Visualize sequencing depth and **ASV abundance distributions**

Environmental DNA (eDNA)

- Genetic material **shed by organisms** into their surroundings
- Can be recovered from **water, soil, sediment** etc.
- Analyzed using molecular methods (e.g. PCR, DNA sequencing, metabarcoding)
- Non-invasive and **highly efficient**, can detect cryptic species
- Typically degrades over hours to weeks, usually **reflects local presence**

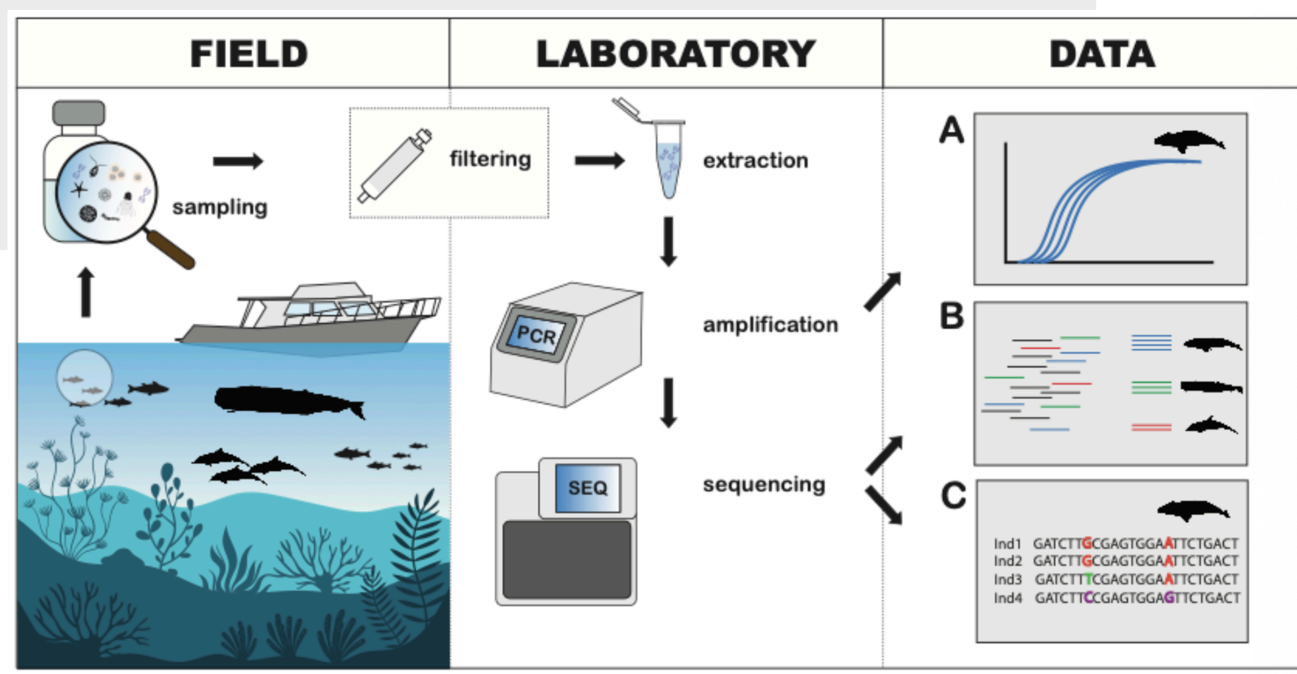


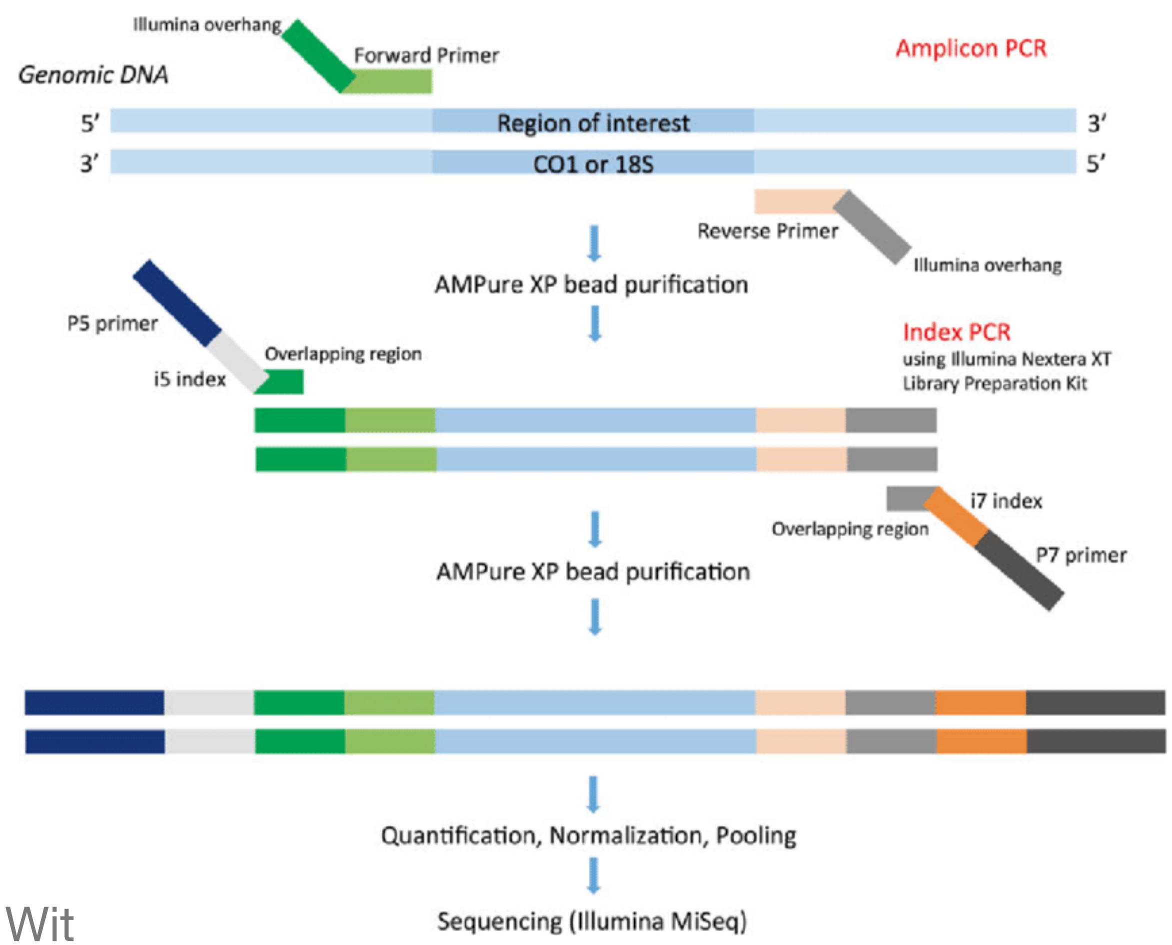
Figure by Székely et al., CC-BY 4.0

Amplicon sequencing

Next-gen sequencing of specific genomic region (e.g. 12S)

Many samples in parallel by **multiplexing** (using unique molecular index for each sample)

Dual-PCR library preparation



Pierre De Wit

Illumina MiSeq bench top sequencer

7–25 million reads, up to 2 × 300 bp

Illumina Inc.

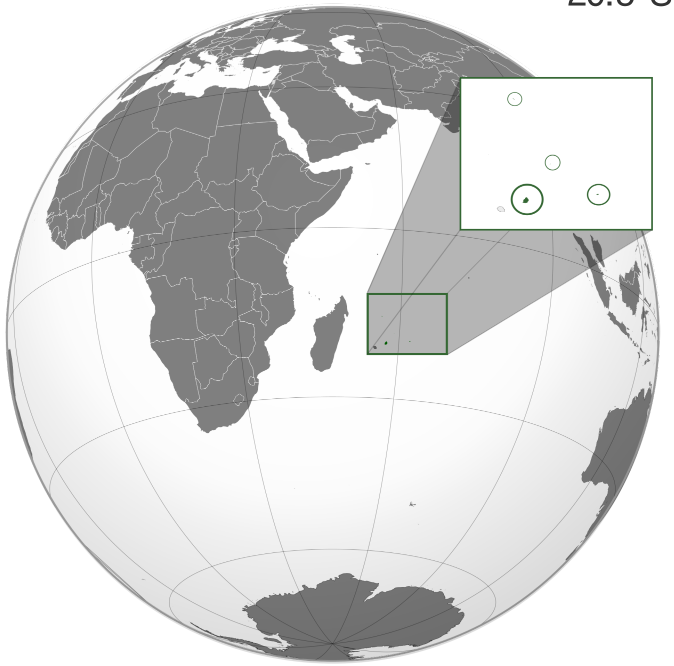
Monitoring seagrass restoration

Dataset

- Restoration began **July 2025**
- Transplantation of single shoots and **plugs** tested at different densities
- eDNA sampling after **0, 3, 6, 12 and 18** months

Kathy Young

Johan Hollander



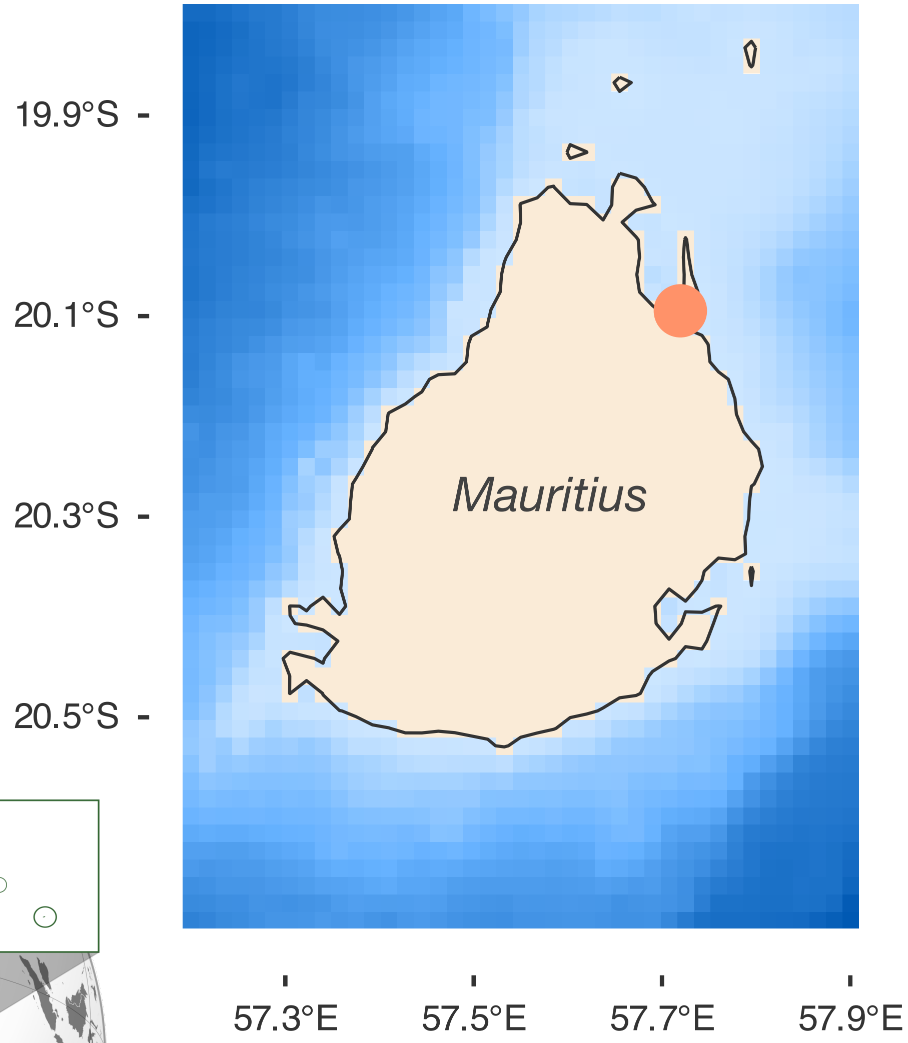


Image by Reef Conservation
Globe by Yashveer Poonit, CC BY-SA 3.0

Monitoring seagrass restoration

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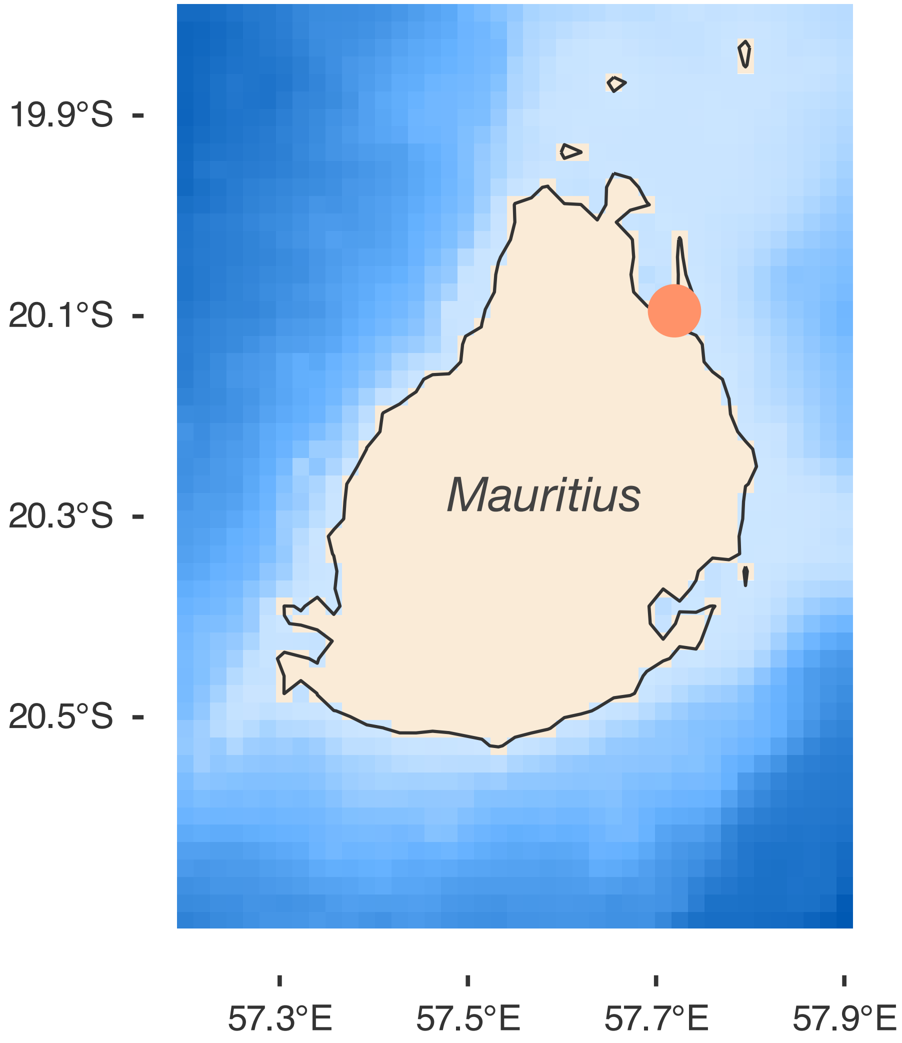


Image by Reef Conservation
Globe by Yashveer Poonit, CC BY-SA 3.0

Sampling scheme and sequencing

Dataset

2 Sites

- Roches Noires
- Bel Ombre

3 Habitats

- Bare sand
- Seagrass meadow
- Channel to open ocean

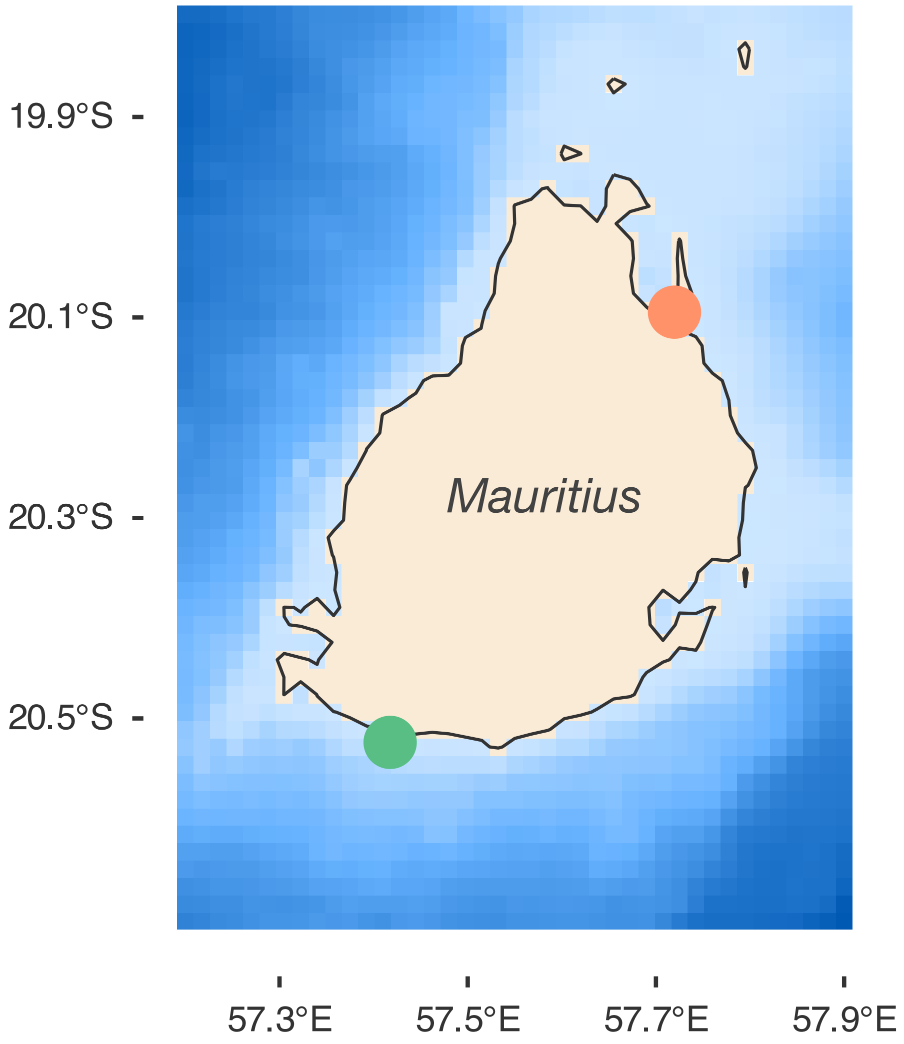
2 Seasons

- summer
- winter

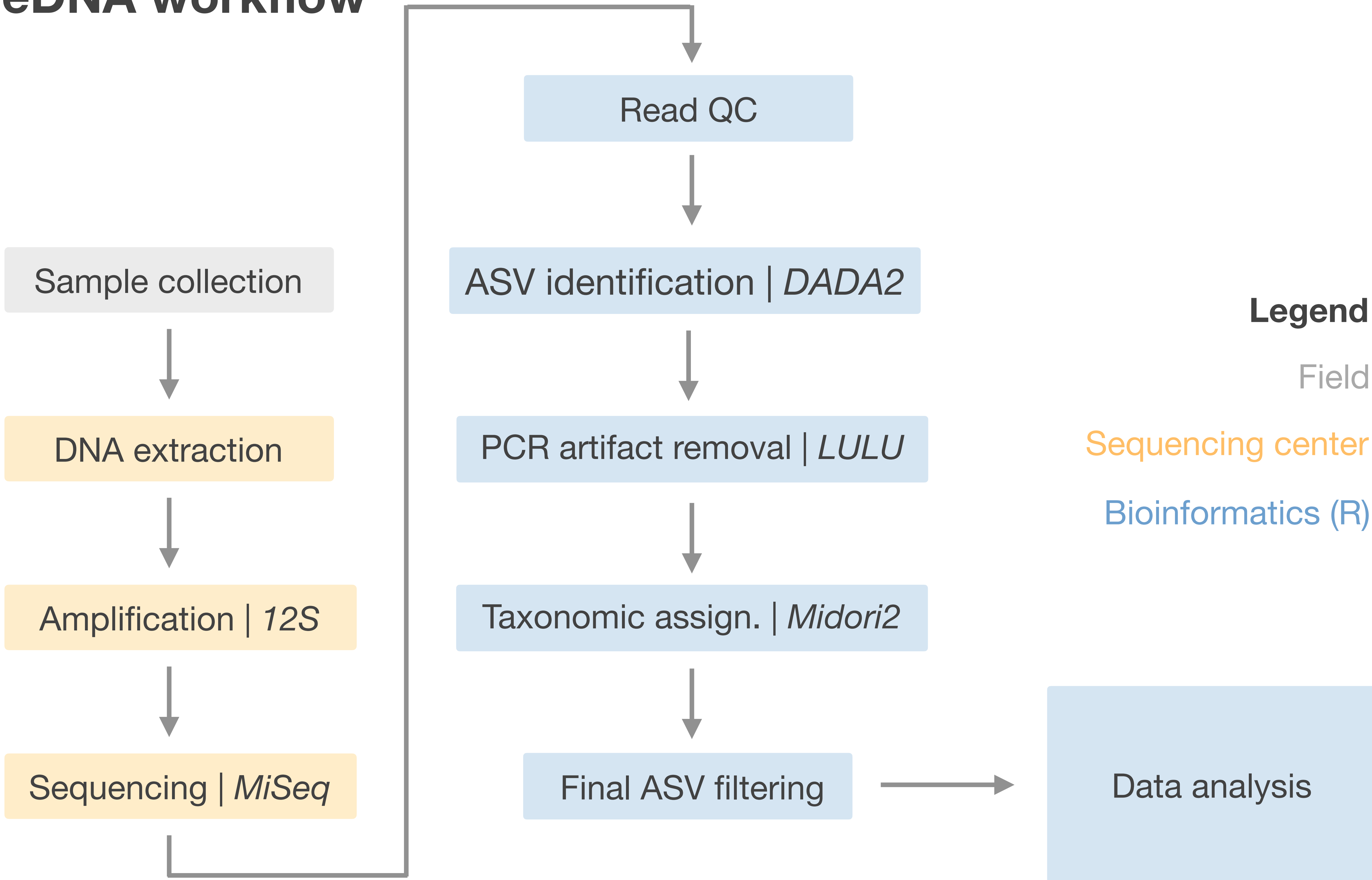
3 Replicates each

Method

- Filtered sea water
- 12S (MiFish)
- Illumina MiSeq



eDNA workflow



Legend

Field

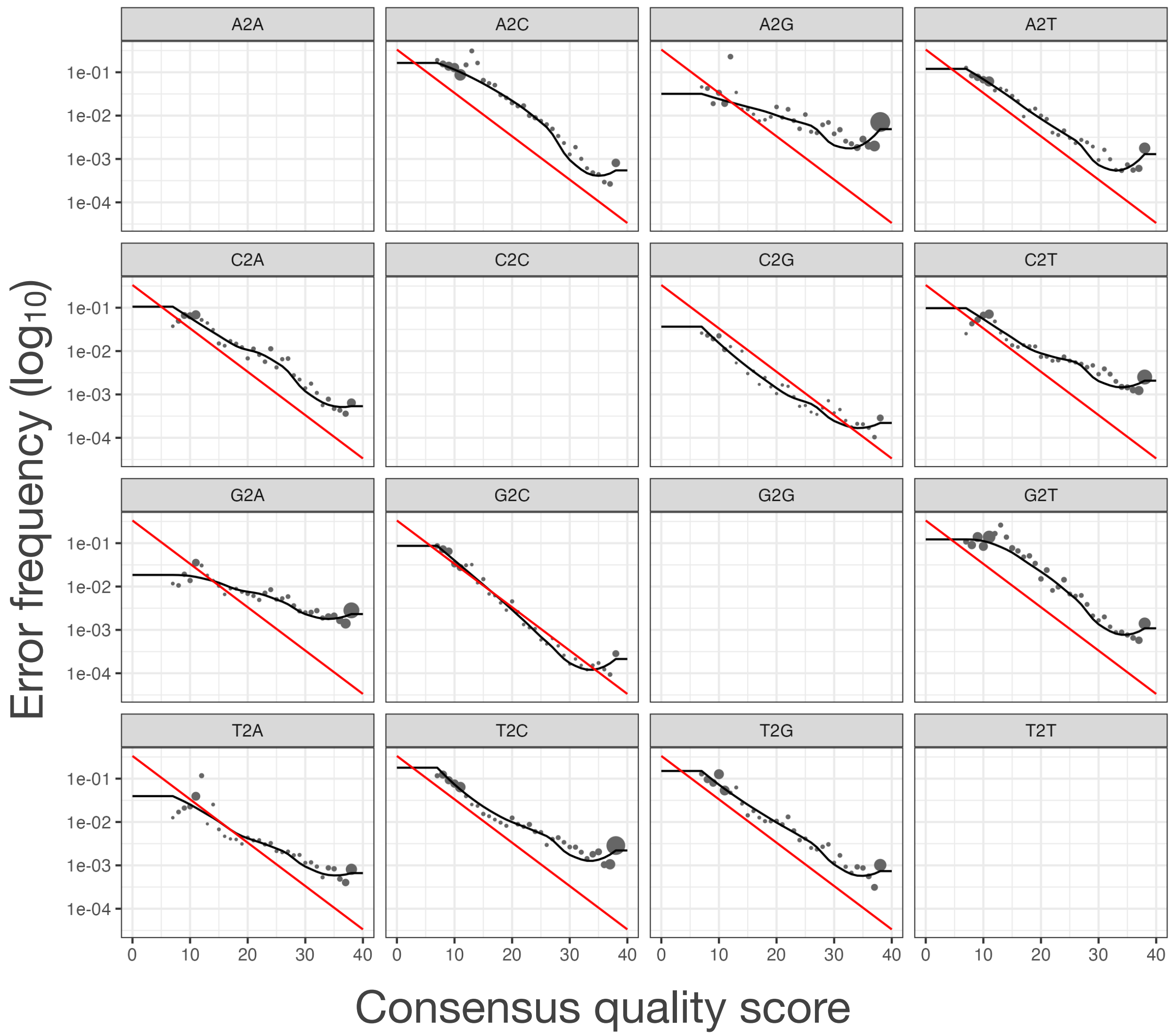
Sequencing center

Bioinformatics (R)

Amplicon Sequence Variant approach

- Error rates are estimated directly from data (black dots)
- Sequences are statistically tested against the error model (black lines)
- Based on probability, sequences are discarded as sequencing errors or retained as true variants (ASVs)
- Better resolution and reproducibility than OTU clustering

Learned error rate model | *DADA2*

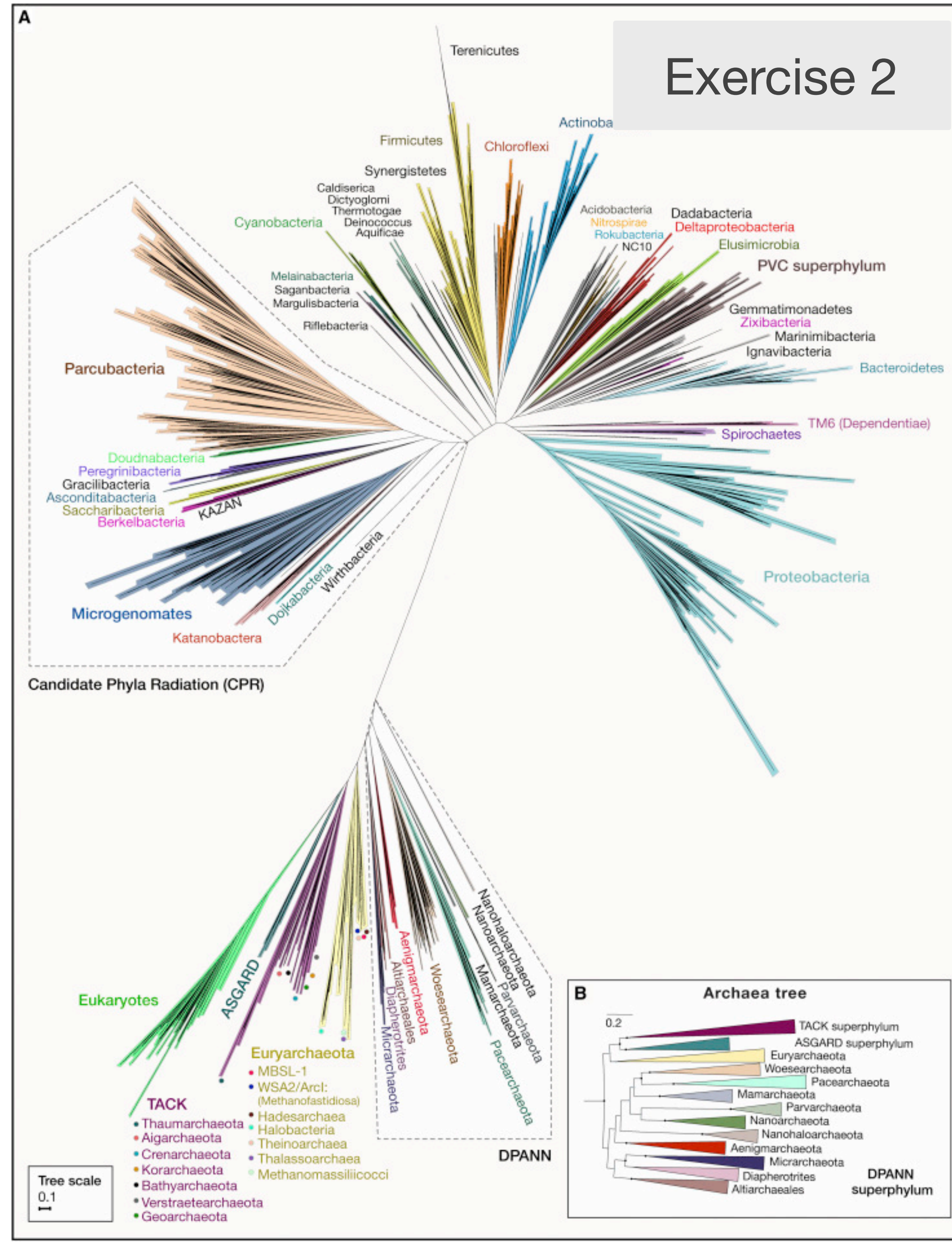


Taxonomic assignment

- Similarity / alignment-based (e.g BLAST)
- Phylogenetic placement
- Probabilistic classifiers (e.g. QIIME2)

high quality ribosomal RNA databases

Exercise 2



Castelle & Banfield 2018, Cell

Matching sequences to database with BLAST

Exercise 2

Algorithm overview

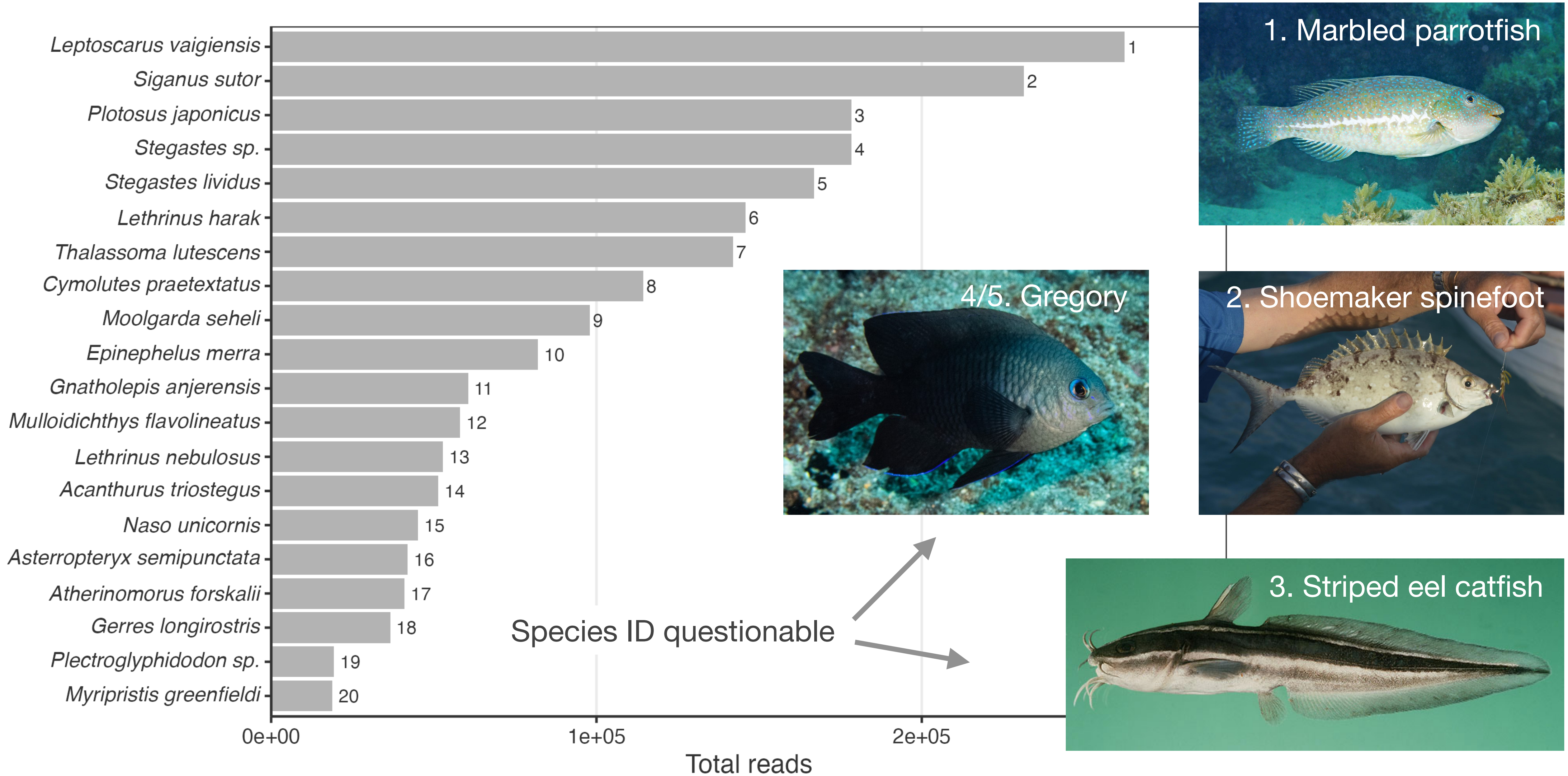
- Split query into very short segments (*k*-mers or words)
- Find exact matches between words and sequences in database (seeds)
- Extend matches to **local alignments** (HSP; stop once too many mismatches occur)
- Evaluate statistical significance of each HSP (**e-value**)

	Description	Scientific Name	Max Score	Total Score	Query Cover	**E value**	Per. Ident	Acc. Len	Accession
☑	Oncorhynchus keta mitochondrial COX1 gene for cytochrome c oxidase subunit 1, partial cds, isolate: OK_M08F	Oncorhynchus keta	1029	1029	100%	0.0	99.64%	772	LC094471.1
☑	Oncorhynchus keta mitochondrial COX1 gene for cytochrome c oxidase subunit 1, partial cds, isolate: OK_M01F	Oncorhynchus keta	1029	1029	100%	0.0	99.64%	772	LC094464.1
☑	Oncorhynchus keta isolate 10_Narva cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Oncorhynchus keta	1029	1029	100%	0.0	99.64%	655	KR778851.1

…

Most abundant species overall

Exercise 4



1. by RJ Koch | 2. by J Travis | 3. by JE Randall (all FishBase), 4/5. by Bill Ludt (NHM LA)

Metabarcoding / eDNA I

Summary

```
### <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< R >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>

ps <- phyloseq(        # Create phyloseq object from (all in suitable format):
  OTU = asv_ps,                                              # ASV count table
  TaxonData = tax_ps,                                    # BLAST taxonomy table
  SampleData = sample_ps                                # sample metadata table
)

ps_filt <- ps %>%                                # remove from phyloseq object:
  prune_samples(sample_sums(.) >= 1000, .) %>%     # samples with < 1000 reads
  subset_taxa(class == "Actinopteri")              # non-(ray-finned)fish ASVs
```

Key takeaways

Summary

- Metabarcoding uses marker genes (e.g. 12S, COI) to identify and compare organisms in complex communities
- Reads can be “denoised” into high resolution **Amplicon Sequence Variants** or more simply clustered into OTUs
- ASV count tables, taxonomic assignments, and sample metadata can be combined into a single structured object (e.g. **phyloseq**) for downstream analysis
- **FItering low-abundance ASVs and low-read samples** is a necessary step before ecological interpretation, since raw ASV tables contain noise and uneven sequencing depth

Please help to evaluate this course

June 26 – July 17